Unit 1.7 The number e   p 51 – 58

Note Title   2015/02/09

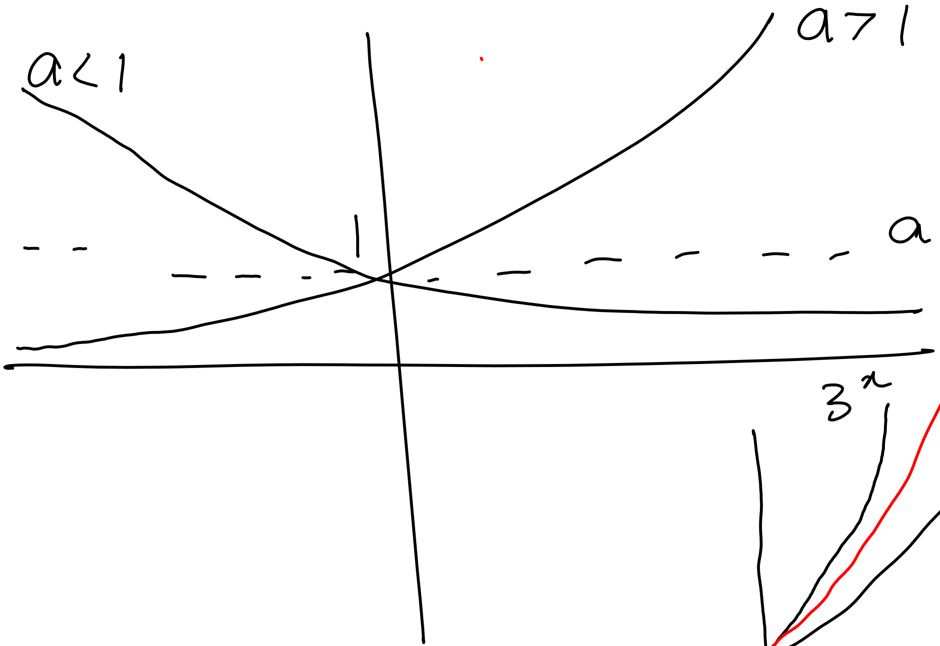
2.71828182845904523536028747135266249775724709369995

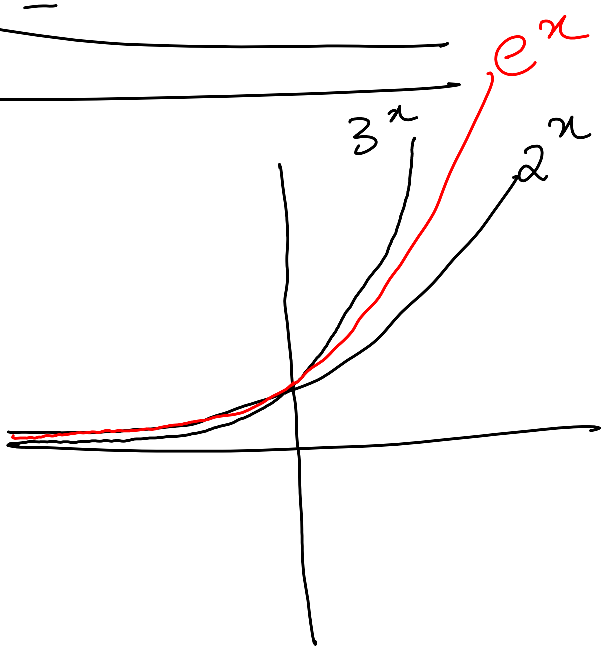
We enjoyed a semester of calculus I reported to everyone

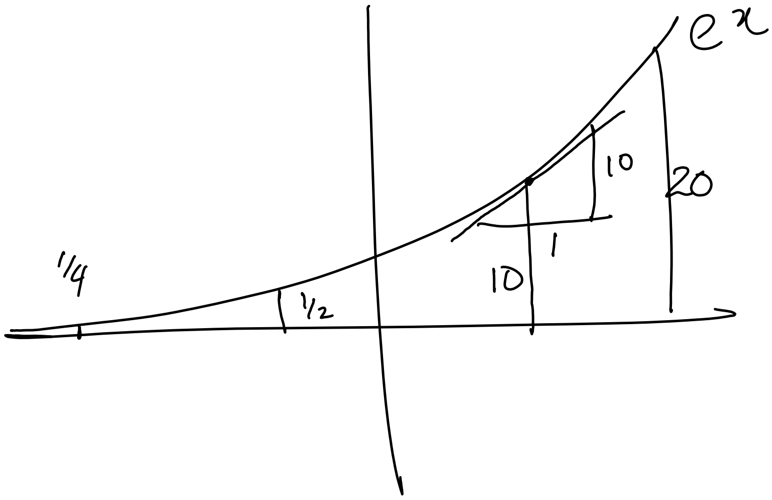
There once was a number called e
for whom it was all about "me"
She thought she was great
But the fact we must debate
We know she wasn't greater than 3

Getting to know more about e   Euler

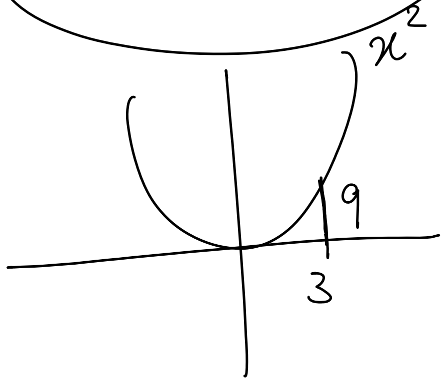
$$y = a^x$$







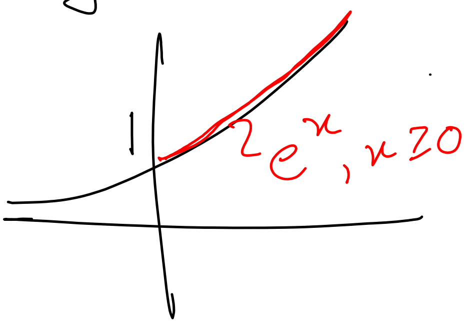
Function value
= gradient value



Eg $y = 100\, e^{0,1 t}$

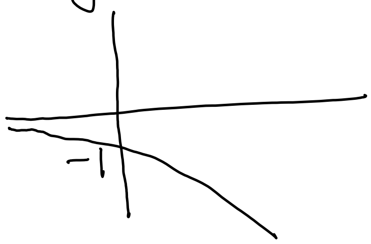
Examples:

1. $y = e^x$

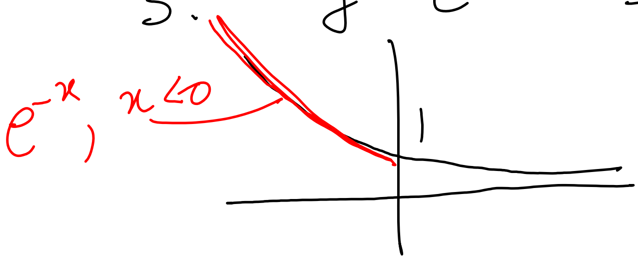


Domain: $\mathbb{R}$
Range: $(0, \infty)$

2. $y = -e^x$

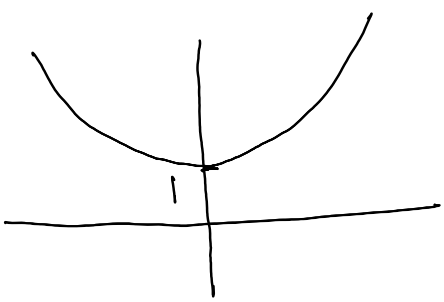


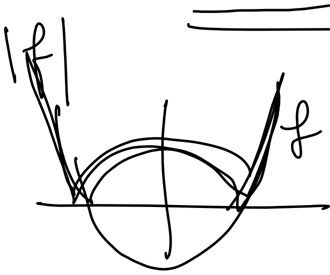
3. $y = e^{-x} = \dfrac{1}{e^x}$



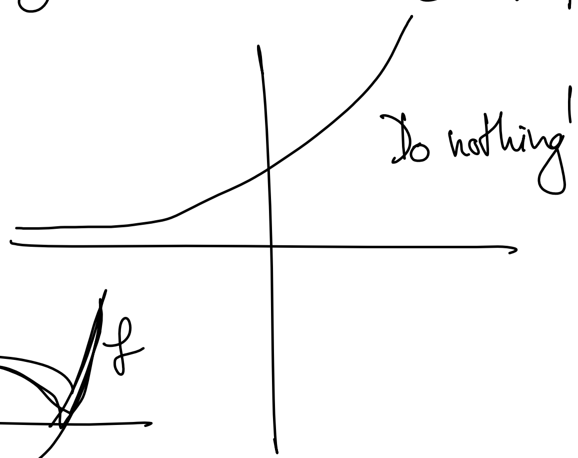
4. $y = e^{|x|}$

$$e^{|x|} = \begin{cases} e^{x} & \text{if } x \geq 0 \\ e^{-x} & \text{if } x < 0 \end{cases}$$

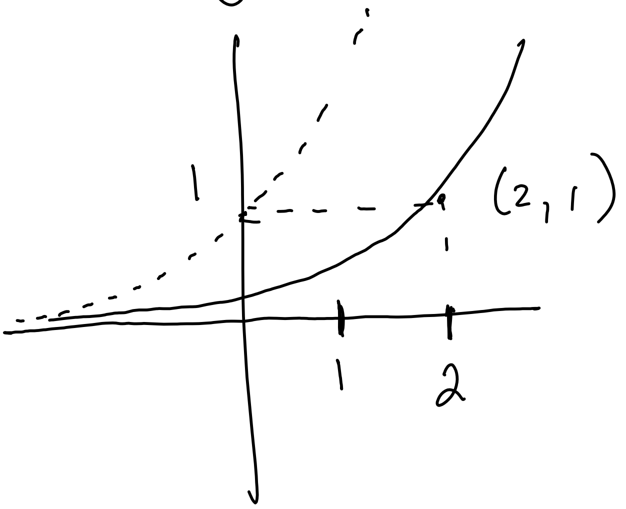




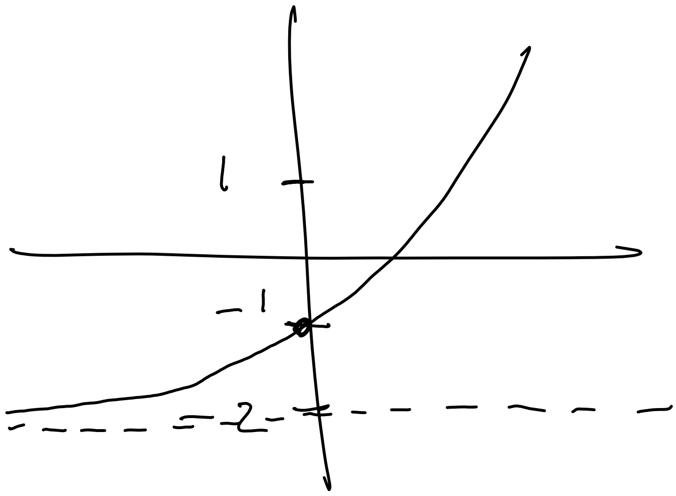
5. $y = |e^x|$



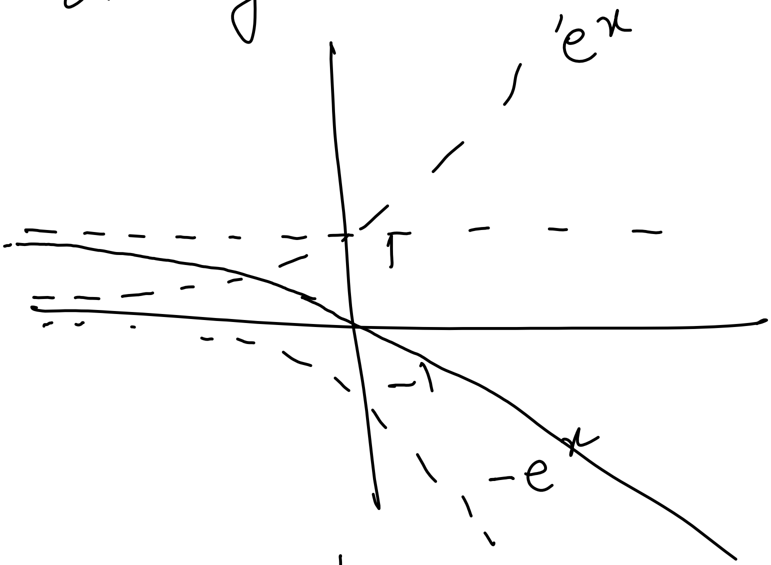
6. $y = e^{x-2}$



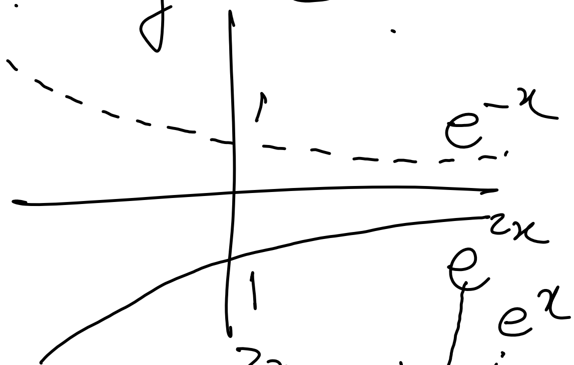
7. $y = e^{x} - 2$



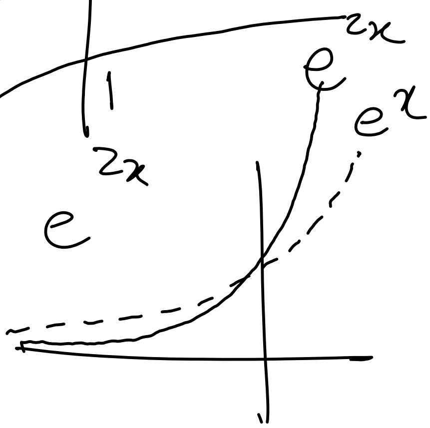
8. $y = 1 - e^{x}$



9. $y = -e^{-x}$



10. $y = e^{2x}$



Exponent laws

1. $e^{a} \cdot e^{b} = e^{a+b}$

2. $\dfrac{e^{a}}{e^{b}} = e^{a-b}$

3. $(e^{a})^{b} = e^{ab}$

$e^{0} = 1$

$e^{-1} = \dfrac{1}{e}$
$= 0{,}37$

$e^{-100} = \dfrac{1}{e^{100}}$

tiny.

Problems Simplify:

$$\frac{e^3\sqrt{e^4}}{e^8} = e^{-3} = \frac{1}{e^3}$$

$$\frac{(e^x . e^y)^4}{(e^x)^y} = e^{4x+4y-xy}$$

$\sqrt{a^2+b^2}$
$\neq a+b$

$\sqrt{e^{2x}+e^{2y}}$ is a simple as possible